Data availability

Chemical
pH (1:5)
Macronutrients (N,Ca, P...)
Micronutrients (Fe,Mb,Cu...)

Biological
Organic Matter (%)
C/N Ratio
Organic carbon (%)

Physical
Clay (%)
Silt (%)
Sand (%)
Texture

Risks

First, administrative load: small farms may find compliance and claims management onerous. SoilHub and advisory services can mitigate this by offering templates, digital workflows and help-desk functions that reduce the time cost of doing the right thing. Second, climate variability: rotations and cover crops must remain viable in dry or cold springs; agronomic menus should therefore include hardier mixes and flexible establishment windows. Third, data fragmentation: if soil-map layers, farm records and agency systems don't talk to each other, transaction costs rise and trust erodes. The remedy is interoperability and light data standards so that evidence for MRV is portable across tools and programmes.

Preliminary results

Scaled across METK's network and adopters, the package should improve aggregate stability, reduce erosion risk on vulnerable textures, and maintain or build soil organic carbon through more frequent living cover and prudent nutrient management. Economically, it supports yield stability and the reputation premium associated with certified seed produced under verifiable soil-friendly systems. Strategically, Estonia demonstrates how a robust formal architecture can be turned into farmer-centred practice when (i) supports are predictable, (ii) MRV piggybacks on existing data, and (iii) a boundary organisation keeps institutions and farms aligned.

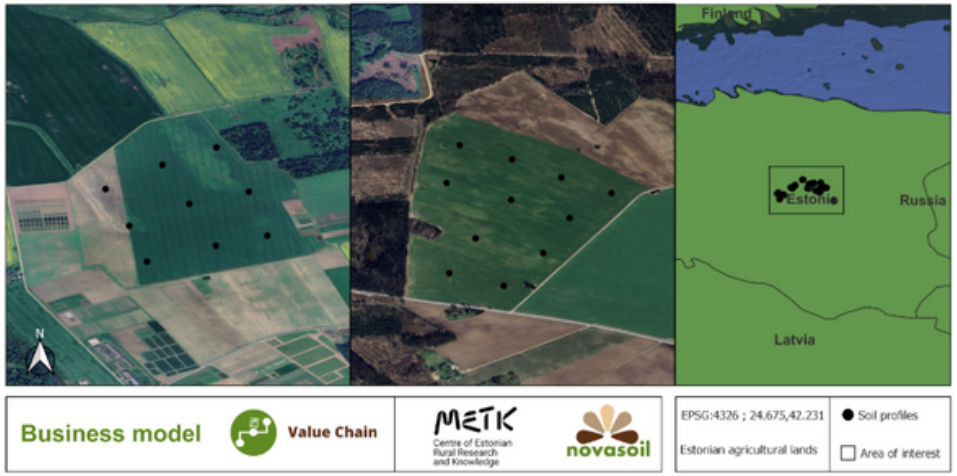
Economic perspective

On the income side, farms combine basic income support with the environment-friendly production package. These pillars de-risk rotations, cover and careful fertilisation, which in turn safeguard the quality and consistency required for certified seed markets. Where field infrastructure limits performance, melioration support finances essential works. Over time, marketing labels and digital transparency (map-based profiles of soil-friendly management) can yield a modest but real premium, particularly in B2B seed channels sensitive to provenance and stewardship

Policy and governance

Estonia's policy environment on soil health relies primarily on supportive incentives—subsidies, compensatory payments and some voluntary finance mechanisms—rather than strict command-and-control rules. This pragmatic, farm-centred approach exerts medium-to-high influence on soil strategies. The governance landscape is formally strong, with ministries, agencies, advisory services and research actors actively involved. However, administrative burdens create friction, particularly for smaller farms that struggle to meet policy requirements. A key innovation is SoilHub (2024), conceived as a boundary organisation to bridge policy, research and land users, helping to align formal ambitions with practical implementation.

Integrated production in Estonia



novasoil

The aim of the NOVASOIL project is to demonstrate that investing in soil health delivers benefits for society and the environment.
The main output of the project is a tool that allows analysis of different soil health investment opportunities across Europe.
This tool is based on case studies in European and third countries that highlight both the uniqueness and the diversity of soil health investments.

Landscape Enterprise Networks

Summarize

The Agricultural Research Centre (METK) is Estonia's agricultural research and knowledge centre under the Ministry of Rural Affairs. METK develops and tests agricultural technologies, produces seeds, researches plant health, and develops agronomic and agro-technological solutions to improve soil health, plant productivity and quality. Based on soil research and monitoring, METK develops solutions to increase soil health and provides agricultural advisory services.



The described activities are related to improving soil health and sustainable farming practices, such as assessing soil organic matter content, conducting soil analyses, and modelling soil fertility using GIS-based methods. The aim is to ensure soil health knowledge transfer, to raise awareness, and to underline the importance of soil health in food production, while ensuring sustainability and high-quality plant production.

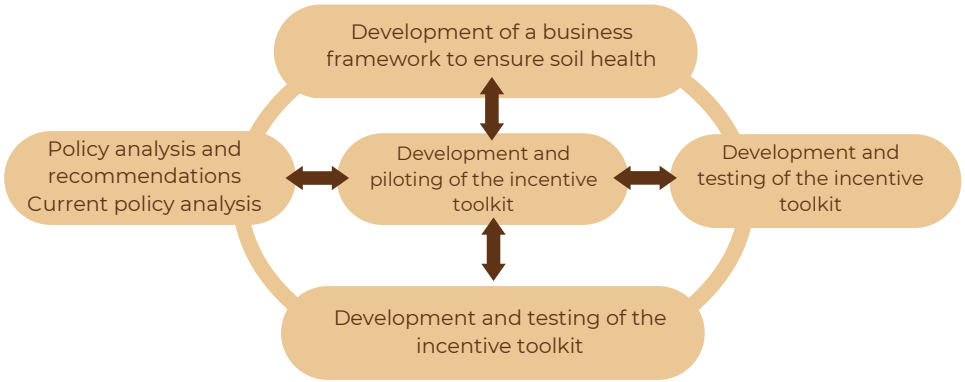


Figure 1. NOVASOIL project structure

Objectives

IIn line with the objectives of the NOVASOIL project, we assess the strengths, weaknesses, opportunities and threats related to the introduction of LENs in order to:

1. encourage farmers/agricultural cooperatives/land managers to adopt soil management practices;
2. motivate corporations/governments to take a holistic, landscape-level perspective on soil health and to commit to improving soil health outcomes.

Case Study Description

Possible inhibitors

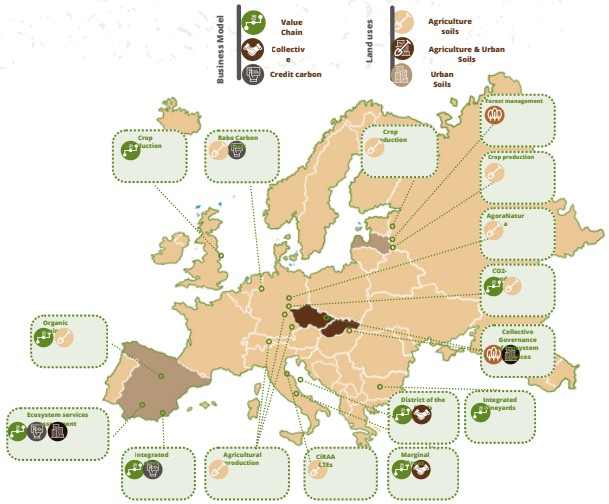
Agriculture faces economic, climatic and technological challenges, including high input costs, lack of support, shortage of skilled labour, climate change impacts, pests and IT risks. In addition, there are problems with the availability of skilled labour and IT security.

Potential enablers

The presence of official and non-official networks, organisations representing farmers, researchers and advisory services, as well as the increasing role of agricultural producers' organisations in the European Union and the growing awareness of the importance of soil health, all contribute to this. Moreover, the EU's long-term vision for rural areas, which extends to 2050, provides a supportive framework.

Possible stakeholders

Other institutions: METK; Ministry of Rural Affairs; Ministry of the Environment; Tallinn University of Technology (TalTech); Estonian University of Life Sciences; University of Tartu; Estonian Chamber of Agriculture and Commerce; NGOs; Scandagra; Baltic Agro; Estonia OÜ; Voore Farm OÜ; Agrone and interested producers.





Specific objectives

SO1 Develop a multidimensional portfolio of business cases and best practices for investing in soil health in Europe and beyond.

SO2 Analyse existing global models and create new incentives and benefits from healthy soils.

SO3 Apply a co-development model for soil health modelling

SO4 Develop prototypes and a test network for incentives.

SO5 Analyse current policies and technologies related to soil health and farmers' incomes.

SO6 Developing a community of practice around investors and key soil health actors.

SO7 Increase interest in soil health through a digital marketing strategy plan.

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